Rotation Matrices Part 2 and intro to HT Matrices

EGR 445/545

GVSU

Plan for today

- review:
  - LA 1 debrief
  - rotation signs and conventions
  - unit vectors as columns
  - dot products
- multiple rotations
- intro to HT matrices
- project 1 kick-off
  - two-link inverse kinematics
- LA 4 and 5

Announcements

- HW 1 is up (github)
    - dues Wed. 5/27
- Quiz 1 next Wed., 5/27
    - what is a rotation matrix?
    - where do rotation matrices come frame?
        - how do I find an expression for one based on a sketch?
    - what does it mean to say the unit vectors are the columns?
- Chapter 2 Learning Outcomes posted

Review

LA 1 Debrief

- what do I mean by verification and why is it important?

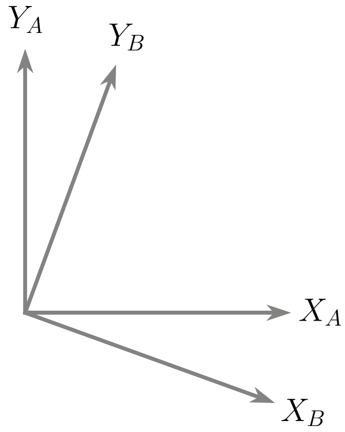
Conceptual Review

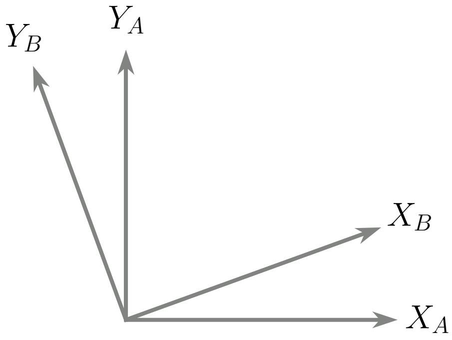
- why do we need to learn rotation matrices?
- why do we also need HT matrices?

Steps and Sign Conventions

- what are the steps and sign conventions in finding ${}^{A}_{B}\mathbf{R}$?

Which of these is a positive rotation?





Dot Products

- what is the relationship between unit vectors and rotation matrices?

Finding ${}^{A}_{B}\mathbf{R} = \mathbf{R}_y(\theta)$

- what sketch do we need to draw?
  - what does a positive $\mathbf{R}_y$ rotation look like?
- which unit vectors do we need to draw?
  - the unit vectors map from which frame to which frame?

Multiple Rotations: the Notation is Your Friend

- step 1: first, we rotate frame 1 from frame 0 by θ_1
- step 2: then, we rotate frame 2 from frame 1 by θ_2
- if we know ${}^2P$, how do we find ${}^0P$?

Finding R with multiple rotations

- step 1: first, we rotate frame 1 from frame 0 by θ_1
- step 2: then, we rotate frame 2 from frame 1 by θ_2
- how do we find ${}^0_2\mathbf{R}$?

$$ {}^0_1\mathbf{R} \cdot {}^1_2\mathbf{R} = ? $$

$$ {}^1_2\mathbf{R} \cdot {}^0_1\mathbf{R} = ? $$

Multiple Rotations and Order of Rotations

Let the notation guide you

Key question:

- If we were going to rotate first from A to B and then from B to C, how would we find ${}^{A}_{C}\mathbf{R}$?
  - can our notation conventions help guide us?
  - what does the math behind ${}^{A}_{C}\mathbf{R}$ really mean?

Remaining Chapter 2 Topics

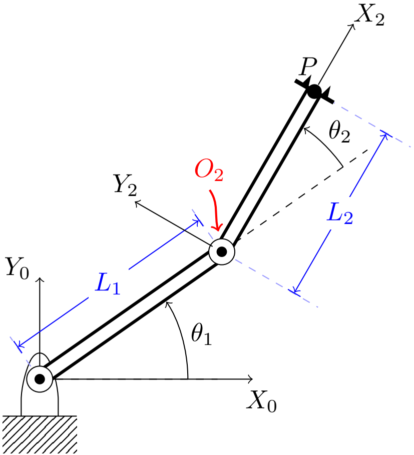
- how do we handle rotation and translation together?
- what is an HT matrix and how do they work?
- how do we find the inverse of an HT matrix?
- do rotation matrices really have 9 unknowns?
- constraints and Euler angles
- what are free vectors and how to we deal with them?
- computational considerations

Intro to HT Matrices

Intro to HT Matrices

- How do we find ${}^{1}P_{tip}$?
- what is wrong with this?:

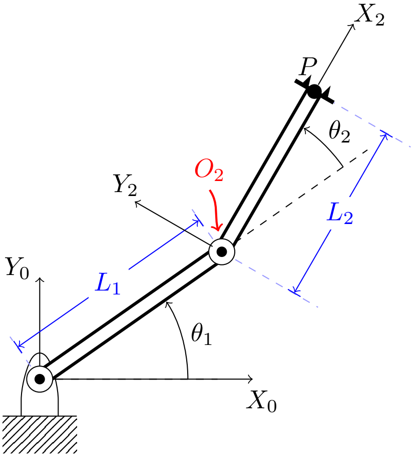
$${}^{1}P_{tip} = {}^{1}_{2}\mathbf{R} \cdot {}^{2}P_{tip}$$



The point of an HT matrix

- how do we put this in one matrix operation?

$$^{1}P_{tip} = {}^{1}P_{O2} + {}^{1}_{2}\mathbf{R} \cdot {}^{2}P_{tip}$$



The Form of an HT Matrix

- how do we add a translation and do a rotation in one step?

$$ {}^{0}_{1}\mathbf{T} = ? $$

Project 1 Kick-Off

- projects folder under student github folder

Two-Link Inverse Kinematics

- under class folders

Active Learning

- learning activities 4 and 5